

Assignment 0

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Setup

```
using Distributions, Plots, LaTeXStrings, LinearAlgebra, Statistics, Random
```

Q1

Install Julia following instructions in lecture notes/etc. and clone this repository so you can open the notebooks in this repository

Q2

Edit the markdown cell below to add some math text with the Pythagorean theorem (e.g. $x^2 + y^2 = z^2$)

(double click to enter your answer here)

Q3

Create a function which takes the sides of the rectangle and calculates the hypotenuse (i.e., code up $z(x, y) = \sqrt{x^2 + y^2}$) and calculate $z(3, 4)$

```
# Modify your code here.
```

Q4

Plot $f(x) = x^2$ for a grid of 20 points of $x \in [0, 1]$. You can make a grid with `range(0, 1, 20)`

```
# Modify your code here.
```

Q3

Write a function that draws a 100 random normal variables (use `randn(100)`) and manually calculates the mean and variance. By manually I mean using sums/etc. to implement the standard formulas for mean and variance of your choosing.

```
# Modify your code here.
```